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title: “Ask the Sensors: Grounded, Explainable Activity Question Answering over Wearable Motion Signals”
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--- *** A deterministic parser built first (spoken-form normalisation, typo
correction, time patterns, ordered operation patterns; 0.3 ms per question) is kept as a baseline
and used where an experiment should isolate the sensor pipeline (Sections 5.3, 5.4 and 6).

3.6 Operations and the grounding rule

Each operation reads the timeline and fills the six required output fields:

Operation	Answer	Evidence cited
identify	activity with the most time in the range	its longest intervals in the range
verify	Yes / Likely yes / No	the matching bouts; for “No”, the dominant activity instead
duration	total seconds	every contributing interval
count	number of bouts ≥ 10 s	the bouts
onset	“Yes, X began at t s”	the first bout, contrasted with the preceding interval
compare	the activity with more (or less) time	both totals
longest	length and span of the longest bout	that bout
prolonged	Yes/No against the 600 s threshold	the longest bout
open	verdict (Consistent / Partially / Not consistent)	intervals of supporting activities

Operations. All numbers are read off the timeline.

The modality field is “Accelerometer, Gyroscope” when the recording has a gyroscope and “Accelerometer” otherwise. Explanations are templates filled with the mean feature values of the cited windows, contrasted with the rest of the recording, for example: *“The cited windows show accelerometer magnitude variability of 0.193 g, a regular rhythm at 2.14 Hz (about 128 cycles/min, periodicity 0.39), gyroscope rotation averaging 0.97 rad/s … For contrast, the rest of the recording averages 0.062 g variability.”* A “Yes” whose supporting intervals have a mean probability below 0.6 is hedged as “Likely yes”, and a “No” is still grounded by citing what the user was doing instead. The language model then rewords this template in plain sentences from the computed fields only; its text is kept only if every number in it appears in those fields (otherwise the template is used: 14 of 670 explanations), and “N/A” answers are never reworded.

3.7 Open-world reasoning

Each behaviour outside the seven labels is described by the recognised activities that would support it and a signal property it requires: exercise needs running, cycling or walking at moderate-to-vigorous intensity; commuting needs sustained walking or cycling bouts (≥ 5 min); sleeping needs lying down with almost no motion; desk work needs still sitting; chores and shopping need light standing-and-moving and walking. The verdict follows from the share of analysed time that meets both conditions (≥ 50% consistent, ≥ 15% partially), and the explanation states that share, the intensity and the longest bout. Falls are handled separately: a fall needs an impact (peak > 2 g with jerk far above the recording’s typical level) ****followed by**** a fall signature — stillness (< 0.03 g variability) or a device-orientation change above 45° within the next minute. Impacts during continuing movement with unchanged orientation are reported as likely footstrikes or bumps, with their timestamps, rather than as falls.

4. Implementation and reproducibility

The code is a Python package (`src/ats/`) with one module per layer, plus scripts:

Step	Command	Output
—	—	—
Download raw data (~16 GB zipped)	`scripts/download_data.py`	`data/raw/` (never committed)
Window features per user	`scripts/build_dataset.py`	`data/processed/`
Train, Figure 2	`scripts/train.py`	`models/hgb_full.joblib`
Tune the decoder (optional)	`scripts/tune_smoothing.py`	tuning table and figure
QA evaluation, Figures 1, 3, 5	`scripts/evaluate.py –robustness`	metrics, records, figures
Efficiency, Figure 4	`scripts/benchmark.py`	`efficiency.json`, figure
Parser evaluation	`scripts/eval_parser.py`	intent accuracy per set
Explanation rubric	`scripts/grade_explanations.py sheet` / `agree`	grading sheets, agreement
Answer questions	`scripts/answer.py –recording r.csv –questions q.txt [–slm Qwen3-0.6B]`	6-field answers
Web page	`streamlit run app.py`	upload, choose model, ask

All randomness is seeded: the user split is stored in `configs/split.json`, question sampling is seeded by a CRC32 hash of the recording id (Python's `hash()` is salted per process), and models use fixed random states. A test suite of 25 checks (`pytest`) exercises every layer on synthetic signals, which are used only for testing and never reported as results. The runnable system reads any CSV with a time column and accelerometer (optionally gyroscope) columns, with loose column-name matching; a 4.4-hour recording is processed in under 1 s, after which each question takes 2–5 s, almost all of it the language model. A Streamlit page (`app.py`) offers the same pipeline: upload one or more recordings, choose Qwen3-1.7B or Qwen3-0.6B, ask a question and read the six-field answer with the recognised activity timeline.

5. Results

5.1 Recognition backbone

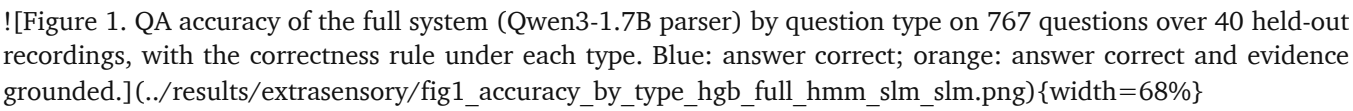
Class	Precision	Recall	F1
—	—:	—:	—:
Lying down	0.49	0.33	0.40
Sitting	0.60	0.58	0.59
Standing in place	0.03	0.06	0.04
Standing and moving	0.10	0.16	0.13
Walking	0.47	0.57	0.52
Running	0.01	0.01	0.01
Bicycling	0.34	0.60	0.43

Per-class window metrics on test users. Macro-F1 0.30 raw, 0.38 after HMM decoding; balanced accuracy 0.33 raw, 0.36 decoded.

Window-level accuracy is modest, and the confusion matrix shows why. (i) **Lying versus sitting**: both are still, so they are told apart only by phone orientation, which depends on where the phone is (pocket, bag, table); 43% of lying windows are predicted as sitting. (ii) **Label noise**: 816 of the 1,087 test windows labelled running come from one user whose phone was essentially motionless (median variability 0.002 g) during those minutes, so no motion-based model can recognise them. Restricted to windows where the phone actually moves (variability ≥ 0.05 g), walking recall is 0.81 and bicycling 0.83. (iii) **Running versus walking**: even on moving windows only 6% of

running is recognised; 76% is called walking, as a phone in a bag does not show the larger impacts a body-worn sensor would. Vaizman et al. score each label one-vs-rest and fuse more sensors (watch, audio, location), so their figures are not directly comparable with a 7-way choice from phone motion alone. For reference, our one-vs-rest balanced accuracies are 0.79 (bicycling), 0.75 (walking), 0.59–0.60 (sitting, lying) and 0.50–0.52 (standing in place, standing and moving, running).

5.2 Question answering

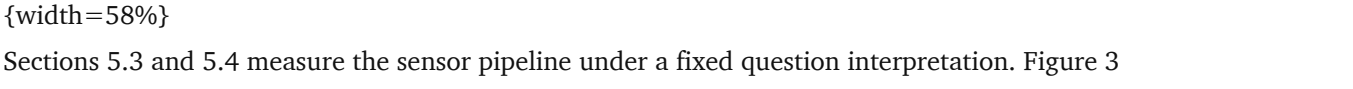


Answer kind	Metric	SLM system
Verification (yes/no, n = 150)	P / R / F1 on “yes”; specificity	**0.85 / 0.73 / 0.79; 0.87**
Prolonged (yes/no, n = 80)	P / R / F1 on “yes”; specificity	**0.70 / 0.92 / 0.79; 0.82**
Onset detected (n = 74)	recall	**0.95**
Identification (4 classes, n = 75)	macro-F1 / balanced accuracy	**0.46 / 0.44**
Comparison (6 winners, n = 80)	macro-F1 / balanced accuracy	**0.70 / 0.74**
Open-world (3 verdicts, n = 80)	macro-F1 / balanced accuracy	**0.67 / 0.69**
Duration (n = 114)	mean / median abs. error; MAPE	**1,107 s / 273 s; 255%**
Count (n = 114)	mean / median abs. error	**4.9 / 1.0 bouts**
Cited evidence (duration, onset)	grounding precision	**0.40**

Answer-level metrics for the SLM system on the same 767 questions.

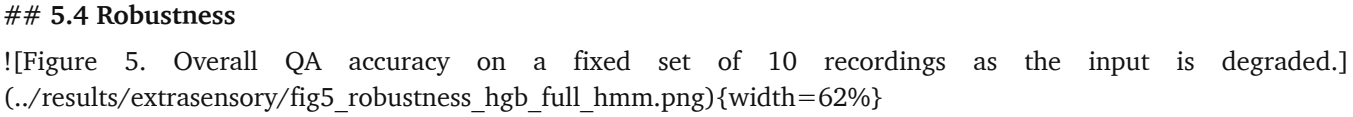
These metrics expose biases that accuracy hides. Verification and prolonged questions remain affected by under-recognised postures and over-merged stationary intervals. Numeric errors are heavy-tailed: posture misclassification in long recordings can substantially increase duration error. With the language model the median duration error rises to 273 s and MAPE to 255%, because a misread time range makes the system measure the wrong stretch of the recording; MAPE is in any case inflated by references of a few minutes.

5.3 Accuracy versus strictness



Sections 5.3 and 5.4 measure the sensor pipeline under a fixed question interpretation. Figure 3 shows whether misses are near or wild. Duration acceptance grows steadily with the tolerance, so many misses are moderate rather than catastrophic. For onsets, the yes/no part is right in 82% of cases; when both system and reference say yes, the predicted onset is within 60 s of the true one in 34% of cases, within 5 minutes in 51% (median error 258 s). The fixed IoU ≥ 0.5 rule is therefore strict for onsets: the first bout is often found a few minutes early or late, or starts at the very beginning of a recording that the reference begins later.

5.4 Robustness



Degradation	Levels → overall QA accuracy
Dropped samples	0%: 0.67 · 10%: 0.66 · 30%: 0.65 · 50%: 0.65 · 70%: 0.63
Sampling rate	25 Hz: 0.67 · 15 Hz: 0.65 · 10 Hz: 0.64 · 5 Hz: 0.61
Gaussian noise	0 g: 0.67 · 0.02 g: 0.62 · 0.05 g: 0.61 · 0.1 g: 0.29 · 0.2 g: 0.28

Robustness (HMM decoder). Recordings are re-resampled from the surviving samples, as the loader would treat a sparser stream.

The system degrades gracefully with missing data and low sampling rates: 70% dropped samples cost 0.04, and sampling at 5 Hz costs 0.06. Gap-aware resampling and the per-window coverage rule make missing data a non-issue, and the features that matter (variability, tilt, cadence up to ~2.5 Hz) survive at 5 Hz. Additive noise is the weak point: up to 0.05 g accuracy holds (the HMM decoder raised accuracy at 0.05 g from 0.26 to 0.61), but at 0.1 g and above accuracy collapses, because still postures are separated by variability differences of a few milli-g, which the noise swamps. Typical phone noise is a few milli-g, so the collapse lies outside realistic conditions.

5.5 Parser generalisation

8. Contribution statement

| Member | Contributions |

|—|—|

| Madhav Aggarwal | **Data and preprocessing.** ExtraSensory loader, unit and timestamp cleaning, faulty-sensor exclusion, 25 Hz resampling with gap masking, 6 s windows, user split. |

| Jyotirmoy Konwar | **Language model.** Qwen3 via llama.cpp, intent schema and prompt, time conversion and grounding checks, explanation rewording with number check, question understanding and evaluation. |

| Venu Gopal Battula | **Evaluation.** Generated QA set and scoring, hand-written question sets, answer-level metrics, evaluation runs and Figures 1, 3, 5, error analysis, explanation rubric. |

| Aryan Khaiwal | **Deployment.** Streamlit page (`app.py`) with upload, model choice, six-field answer and timeline; command-line script. |

References

- Y. Vaizman, K. Ellis, G. Lanckriet. Recognizing detailed human context in the wild from smartphones and smartwatches. *IEEE Pervasive Computing* 16(4), 2017.

- J. Ba, R. Caruana. Do deep nets really need to be deep? *NeurIPS*, 2014.

- G. Ke et al. LightGBM: a highly efficient gradient boosting decision tree. *NeurIPS*, 2017 (the histogram-based boosting that scikit-learn's HistGradientBoosting follows).

- L. Rabiner. A tutorial on hidden Markov models and selected applications in speech recognition. *Proceedings of the IEEE* 77(2), 1989.

- Qwen Team. Qwen2.5 technical report, arXiv:2412.15115, 2024; Qwen3 technical report, arXiv:2505.09388, 2025.

- G. Gerganov et al. llama.cpp, https://github.com/ggml-org/llama.cpp (GGUF quantisation of Qwen3 by Unsloth).